# Unit 4

## Solutions
## Acids & Bases

## Solution Definitions

*Homogeneous mixture* – a uniform mixture of only one phase. (ex. salt water, kool-aid)

*Heterogeneous mixture* – a mixture that consists of physically distinct parts. (ex. smoke in air, sugar & salt)

*Solution* – a homogeneous mixture of substances composed of at least one solute and one solvent.

*Solute* – a substance that is dissolved into a solvent. (ex. NaCl)

*Solvent* – the medium in which a solute is dissolved; often the liquid component of a solution. (ex. water)

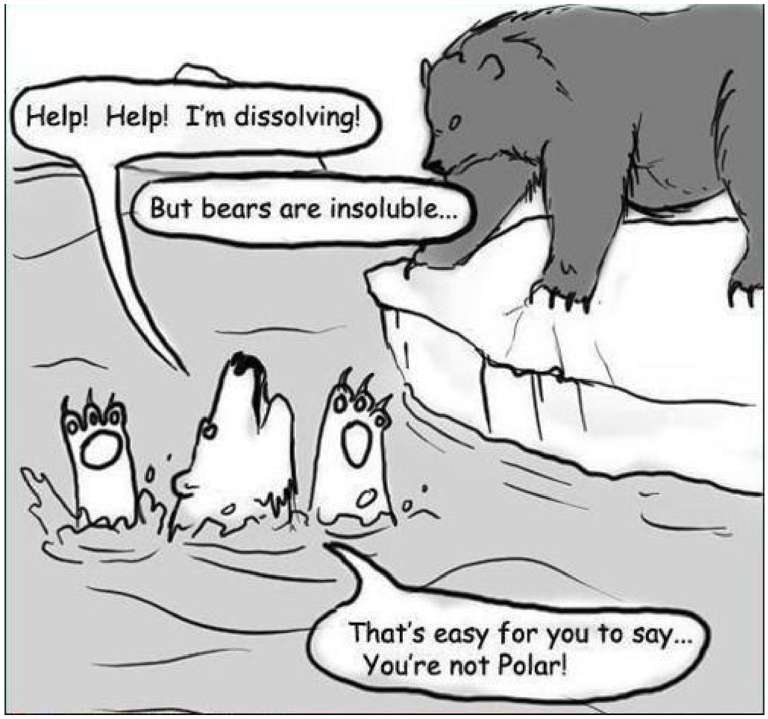
A solution does not always consist of a solid dissolved in a liquid:

| SOLUTE | SOLVENT | EXAMPLE |
| --- | --- | --- |
| gas | gas | Oxygen in air |
| gas | liquid | $CO_2$ in soda |
| gas | liquid | Oxygen in water |
| liquid | liquid | Alcohol in water |
| solid | solid | metal alloy |

Why are some things soluble and some not?

A solute's ability to dissolve in a solvent is dependant on the potential intermolecular forces (DDF, LDF, H-Bonding) than can occur between them.

As a rule-of-thumb, **like dissolves like.**
Polar solvents will dissolve polar solutes. Non-polar solvents will dissolve non-polar solutes.



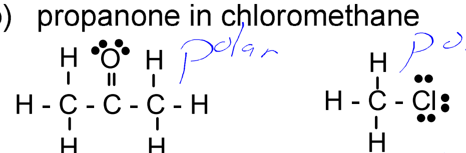

Ex) Is it soluble? Why or why not?

a) ammonia in water

Yes, both are polar

b) propanone in chloromethane

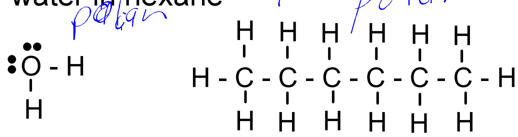
polar polar



Yes

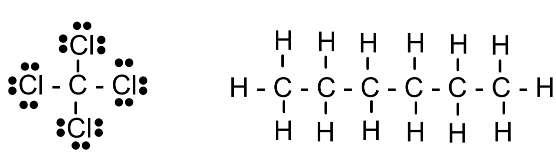
c) water in hexane

polar non-polar



NO

d) carbon tetrachloride in hexane



Yes, both are non-polar

Water is known as the "universal solvent".

Both DDF and H-Bonding occurs in $H_2O$, so it is able to dissolve a number of solutes.

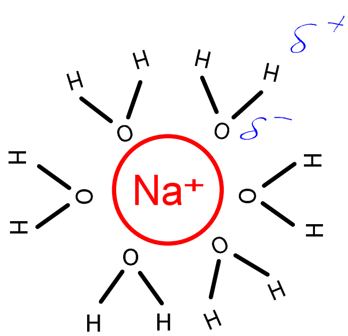
## Recall: Total Ionic Equations

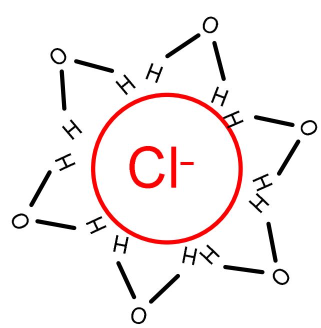
$$NaCl_{(aq)} \longrightarrow Na^+_{(aq)} + Cl^-_{(aq)}$$

Ionic Compounds will dissolve in polar solvents (especially $H_2O$) since they are able to form ion-dipole interactions.

Ions dissociate and are surrounded by water molecules.

For example NaCl in $H_2O$.



http://www.youtube.com/watch?v=xdedxfhcpWo

Read p354-358
Do p358 #1-5, 7, 13